

TriCognia Ville

Overview

TriCognia Ville is a story-based gamified reading application designed to help Grade 7 learners rebuild a faded village by completing reading and comprehension quests. As students master each task — multiple choice, drag-and-drop, read-aloud, fluency, tap-target, and punctuation — buildings gradually return from grayscale to full color, livening the town with characters, lights, and ambient activity. Built as a landscape Android app using Godot 4.6 with a Next.js 16 backend, Neon PostgreSQL, and Groq-powered feedback for adaptive reading support.

Features

Reading Quest Modules

- 6 Interactive Quest Types
 - Multiple Choice: Comprehension and vocabulary questions with immediate feedback
 - Drag-and-Drop: Sentence ordering and word/phrase matching with chip-based UI
 - Read-Aloud: Speech-recognized passage reading with pronunciation scoring
 - Fluency Drill: Timed reading practice tracking words-per-minute and accuracy
 - Tap-Target: In-scene tap challenges for vocabulary and visual comprehension
 - Punctuation Read: Practice expressive reading guided by punctuation cues
- Reading-Level Aware Content matched to Grade 7 difficulty bands
- AI-Powered Feedback via Groq for context-sensitive hints and reflection prompts
- Hint Manager with progressive scaffolding when a learner struggles
- Quest Tracker Panel showing active, available, and completed quests

Village Restoration Gamification

- Grayscale-to-Color Buildings: Each completed quest restores a building to full color via custom canvas-item shader
- Unlock Cutscenes celebrating each restored landmark
- Town Livener System that progressively spawns NPCs, lights, and ambient details as the village awakens

- Town Celebration triggered when major story milestones are reached
- Badge Toast Notifications for achievements and reading milestones
- Stars and Progress Tracking across all quests and buildings
- Tutorial Overlay and Quest Tutorial Guide for first-time learners

Story & Dialogue

- Story Manager driving a narrative arc tied to village restoration
- Dialogue Panel for character interactions and quest framing
- Reflection Engine prompting learners to consolidate what they read
- Onboarding Screen introducing the world, characters, and core mechanic
- Splash Screen with branded boot sequence

Audio & Speech

- Text-to-Speech read-aloud of passages, instructions, and dialogue
- Speech Recognition via the SpeechPlugin Android addon for read-aloud and fluency tasks
- Audio Manager for music, ambient village sounds, and SFX with mic ducking
- Mic-Aware Audio Ducking that lowers BGM and ambient layers while recording
- Punctuation-Sensitive Pacing for expressive reading practice

User Management

- Email and Password Authentication with JOSE-signed JWT sessions
- Bcrypt Password Hashing for secure credential storage
- Auth Screen with sign-in, sign-up, and validation flows
- Profile Panel and Profile Button accessible from the village map
- Player Profile Persistence across sessions and devices via the backend
- Network Gate that gracefully handles offline and reconnect states

Backend & Sync

- Next.js 16 App Router API for accounts, quests, progress, and certificates

- Neon Serverless PostgreSQL with Drizzle ORM schema and migrations
- Sync Manager pushing local progress to the cloud and pulling updates
- Quest Content API serving level-appropriate passages and questions
- Gamification API for stars, badges, and unlock state
- Certificate Generation using pdf-lib for printable completion records
- Cloudinary Integration for asset and image delivery
- Reading-Level Service mapping learner performance to Grade 7 reading bands

UI/UX Design

Design Principles

- Landscape-First Layout: Locked landscape orientation optimized for tablets
- Touch-Friendly Targets: Large hit areas tuned for finger input on classroom devices
- Smooth Animations: Tweened building reveals, dialogue transitions, and HUD pops
- Accessibility-Aware: Text scaling, TTS support, and high-contrast UI elements

Theme System

- Custom Style Factory and Theme Builder for consistent button, panel, and label styling
- Storybook-Inspired Branding: Warm village palette that shifts as the town comes alive
- Custom Typography optimized for sustained reading on tablets
- Consistent Spacing across HUD, dialogue, and quest overlays

Security & Privacy

Data Protection

- Bcrypt Password Hashing: Industry-standard salted password storage
- JWT Session Tokens: Stateless authentication via JOSE
- Authentication Required: All progress and quest endpoints require valid auth
- Input Validation: Zod schemas on every API route, plus client-side checks
- Per-User Data Isolation: Progress, attempts, and certificates scoped to the learner

Privacy Features

- Minimal Data Collection: Only essential learner and progress information
- Local-First Saves: Progress persists on-device and syncs opportunistically
- Secure Communication: HTTPS/TLS for all backend requests
- Account-Scoped Media: Cloudinary assets served via authenticated references

Performance Optimizations

Efficiency Features

- GL Compatibility Renderer: OpenGL ES 3.0 for broad Android tablet support
- Custom Grayscale Shader applied per-sprite to avoid full-viewport post-processing cost
- Autoload Singletons (GameManager, QuestManager, StoryManager, AudioManager, TTSManger) for low-overhead global state
- Lazy Quest Loading: Quest content and assets loaded on demand per building
- Tween-Based Animations instead of per-frame scripts for transitions

Resource Management

- Compressed Texture Imports (ETC2/ASTC) for efficient GPU memory use
- OGG Audio for music and ambient layers
- Pooled Dialogue and Toast Components to avoid runtime allocation churn
- Network Gate Backoff for resilient sync under flaky classroom Wi-Fi

Supported Devices

Minimum Requirements

- Android Version: 7.0 (API level 24)
- Architecture: arm64-v8a
- RAM: 2GB minimum, 4GB recommended
- Storage: ~120MB for app and bundled content
- Network: Internet connection required for sign-in and progress sync
- Audio: Microphone access for read-aloud and fluency drills

Optimal Experience

- Android Version: 10.0+ (API level 29+)
- RAM: 4GB or more
- Display: 10" tablet, 1280×720 or larger, landscape
- Audio: Speakers or headphones for TTS and ambient village audio
- Microphone: Built-in or external mic for pronunciation feedback

Acknowledgments

Technologies Used

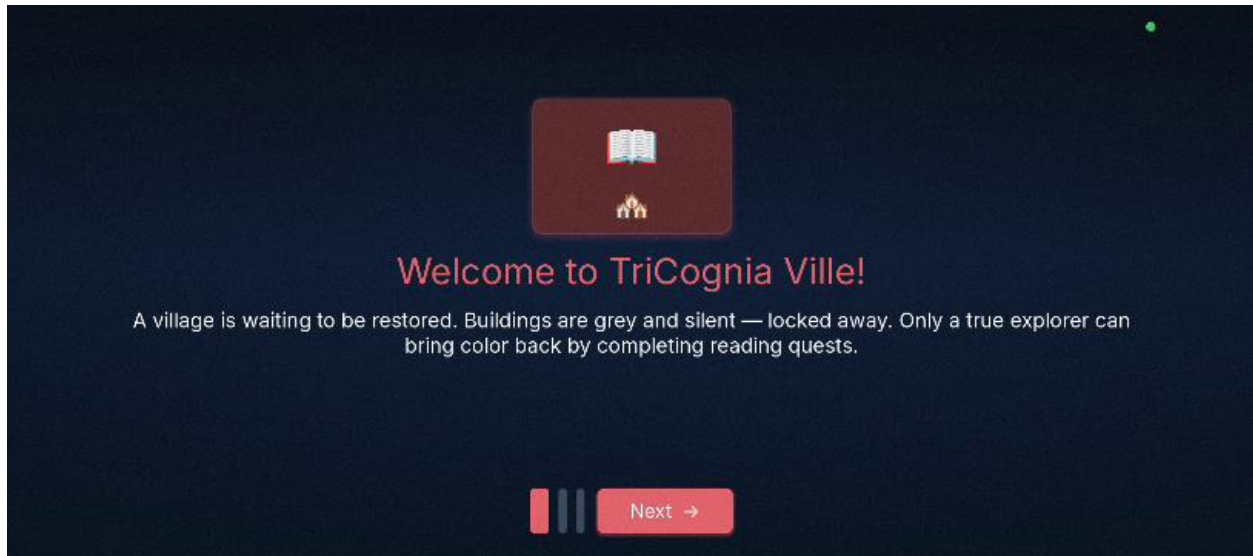
- Godot 4.6 — Game engine with GL Compatibility renderer
- GDScript — Gameplay, UI, and quest logic
- SpeechPlugin — Android speech recognition addon
- Next.js 16 — Backend API with App Router
- React 19 — Admin and dashboard UI
- TypeScript 5 — Type-safe backend code
- Drizzle ORM — Type-safe PostgreSQL access
- Neon PostgreSQL — Serverless database
- Groq SDK — LLM-powered reading feedback
- JOSE + bcryptjs — JWT auth and password hashing
- Zod — API input validation
- pdf-lib — Certificate generation
- Cloudinary — Asset and image delivery
- Tailwind CSS — Utility-first styling for the dashboard

Educational Content

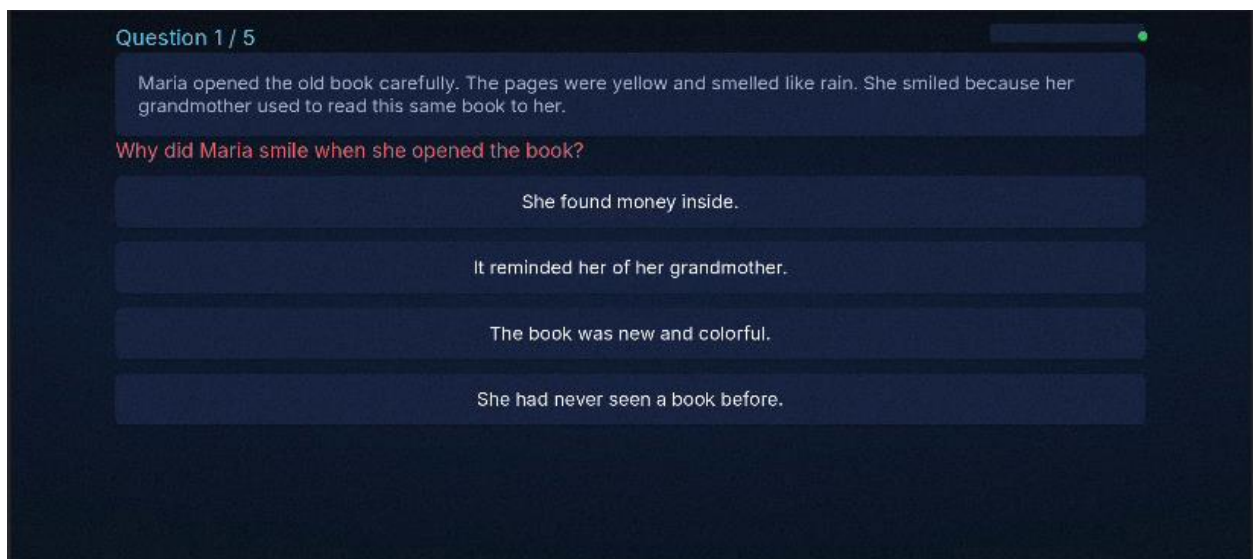
- Reading passages and comprehension items aligned with Grade 7 reading bands
- Story arc and village lore designed to motivate sustained reading practice

- Quest taxonomy spanning literal comprehension, inference, fluency, and expressive reading

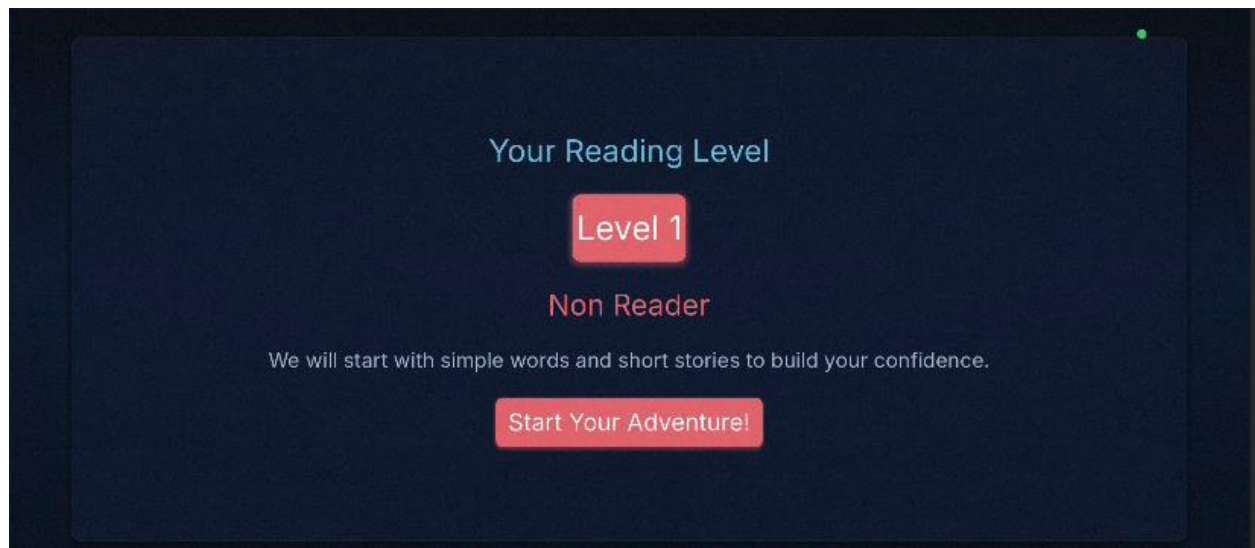
Page Overview



ONBOARDING



ASSESSMENT



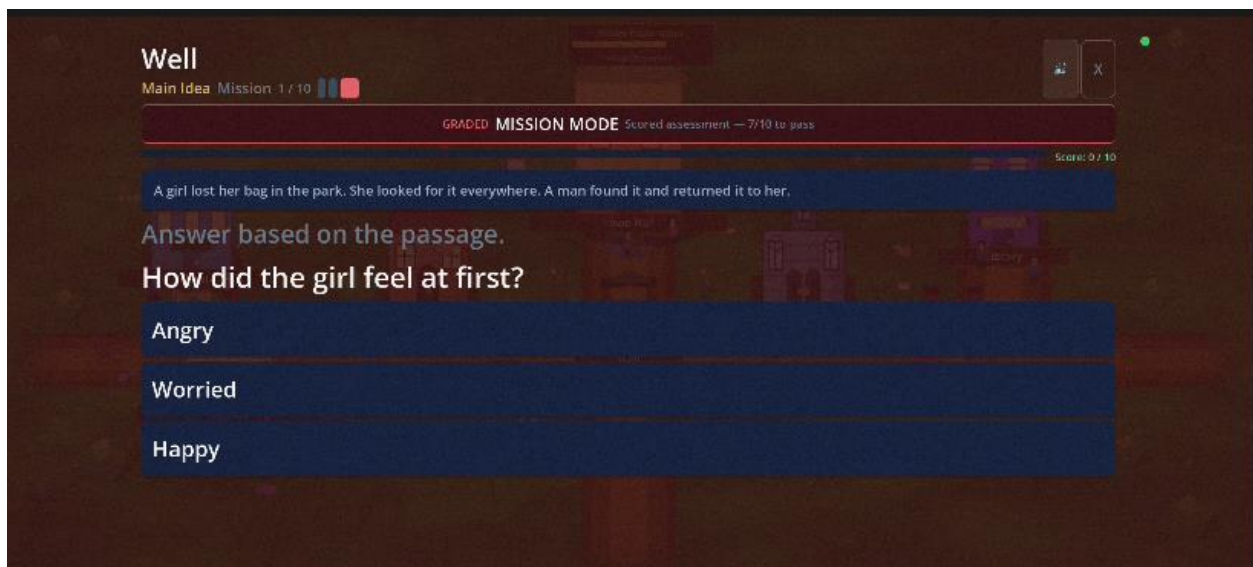
READING LEVEL



VILLAGE (NOT FINISHED)



VILLAGE(FINSIHED)



MISSIONS



PROFILE



DIALOGUES



TASKS